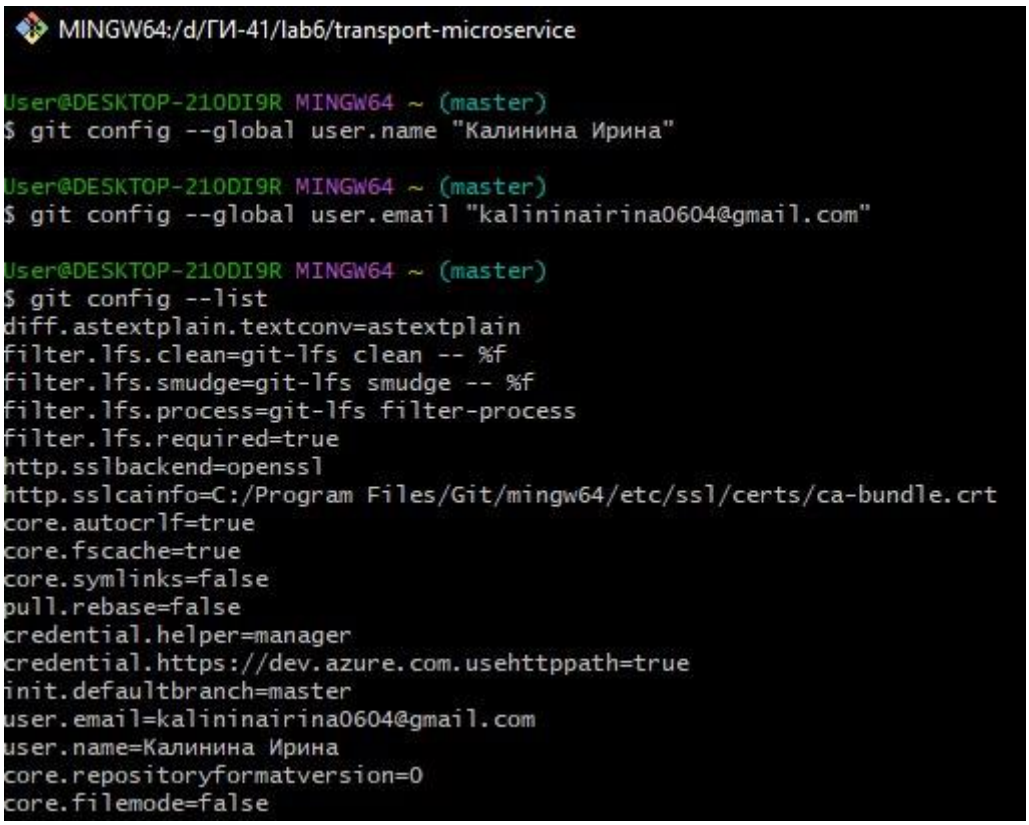

MINGW64:/d/ГИ-41/lab6/transport-microservice
User@DESKTOP-210DI9R MINGW64 ~ (master)
$ git config --global user.name "Калинина Ирина"
User@DESKTOP-210DI9R MINGW64 ~ (master)
$ git config --global user.email "kalininairina0604@gmail.com"
User@DESKTOP-210DI9R MINGW64 ~ (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=kalininairina0604@gmail.com
user.name=Калинина Ирина
core.repositoryformatversion=0
core.filemode=false


User@DESKTOP-210DI9R MINGW64 ~ (master)
$ cd /d/ГИ-41/lab6
User@DESKTOP-210DI9R MINGW64 /d/ГИ-41/lab6
$ git clone https://github.com/KalininaIrina/transport-microservice.git
Cloning into 'transport-microservice'...
warning: You appear to have cloned an empty repository.
User@DESKTOP-210DI9R MINGW64 /d/ГИ-41/lab6
$ cd transport-microservice
User@DESKTOP-210DI9R MINGW64 /d/ГИ-41/lab6/transport-microservice (main)
$ git checkout -b feature/rest-api
Switched to a new branch 'feature/rest-api'
User@DESKTOP-210DI9R MINGW64 /d/ГИ-41/lab6/transport-microservice (feature/rest-api)
$ git branch
User@DESKTOP-210DI9R MINGW64 /d/ГИ-41/lab6/transport-microservice (feature/rest-api)
$

```
User@DESKTOP-210DI9R MINGW64 /d/ГИ-41/lab6/transport-microservice (feature/rest-api)
$
$ #!/bin/bash

# Создаём структуру папок
mkdir -p app
mkdir -p .github/workflows

# Создаём requirements.txt
cat > requirements.txt << 'EOF'
fastapi==0.103.1
uvicorn==0.23.2
pydantic==2.3.0
EOF

# Создаём app/__init__.py
touch app/__init__.py

# Создаём app/models.py
cat > app/models.py << 'EOF'
from pydantic import BaseModel, Field


class RouteBase(BaseModel):
    name: str = Field(..., description="Название маршрута")
    start_point: str = Field(..., description="Точка отправления")
    end_point: str = Field(..., description="Точка прибытия")
    distance_km: float = Field(..., gt=0, description="Расстояние в км")


class RouteCreate(RouteBase):
    pass


class RouteResponse(RouteBase):
    id: int = Field(..., description="Уникальный идентификатор")
EOF

# Создаём app/main.py
cat > app/main.py << 'EOF'
from fastapi import FastAPI, HTTPException, status
from app.models import RouteCreate, RouteResponse

app = FastAPI(
    title="Transport Microservice API",
    description="API для управления маршрутами",
    version="1.0.0"
)

# In-Memory хранилище (заглушка)
routes_db: dict[int, dict] = {}
current_id = 1


@app.get(
    "/api/v1/routes/",
    response_model=list[RouteResponse],
    status_code=status.HTTP_200_OK,
    summary="Получить список всех маршрутов"
)
async def get_routes():
    return list(routes_db.values())


@app.get(
```

MINGW64:/d/ГИ-41/lab6/transport-microservice

```
# Создаём Dockerfile
cat > Dockerfile << 'EOF'
FROM python:3.11-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY ./app ./app

EXPOSE 8000

CMD ["uvicorn", "app.main:app", "--host", "0.0.0.0", "--port", "8000"]
EOF

# Создаём docker-compose.yml
cat > docker-compose.yml << 'EOF'
version: '3.8'

services:
  transport-api:
    build: .
    container_name: transport_service
    ports:
      - "8000:8000"
    restart: unless-stopped
EOF

# Создаём .github/workflows/ci.yml
cat > .github/workflows/ci.yml << 'EOF'
name: CI Pipeline

on:
  push:
    branches: [ "main", "feature/**" ]
  pull_request:
    branches: [ "main" ]

jobs:
  build-and-lint:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v3

      - name: Set up Python
        uses: actions/setup-python@v4
        with:
          python-version: '3.11'

      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install flake8 fastapi uvicorn pydantic

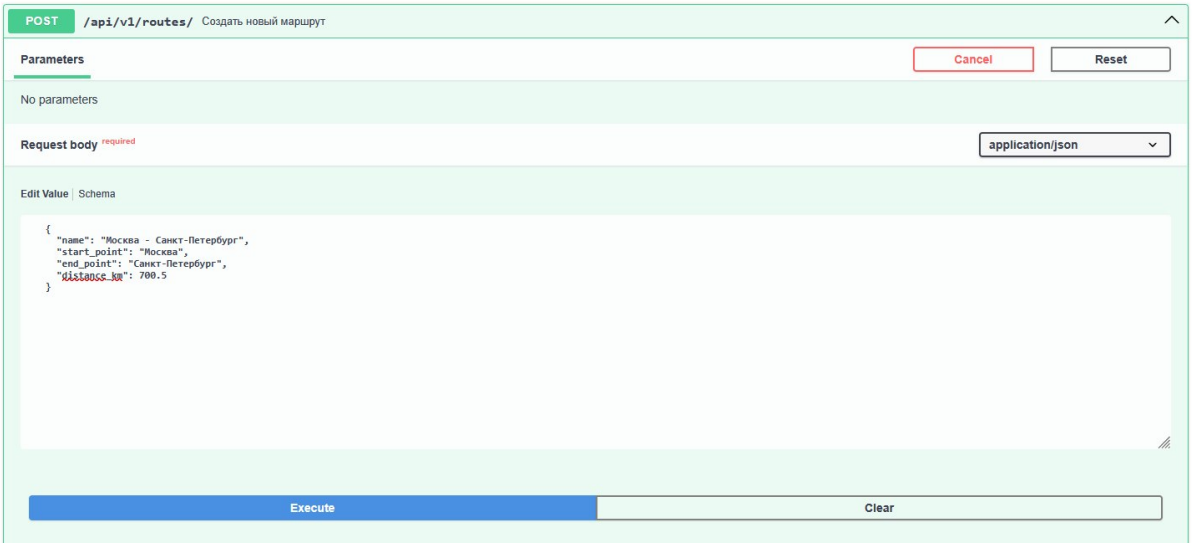
      - name: Lint with flake8
        run: |
          flake8 . --count --select=E9,F63,F7,F82 --show-source --statistics
          flake8 . --count --exit-zero --max-complexity=10 --max-line-length=120 --statistics

      - name: Build Docker Image
        run: docker build -t transport-service:test .
EOF

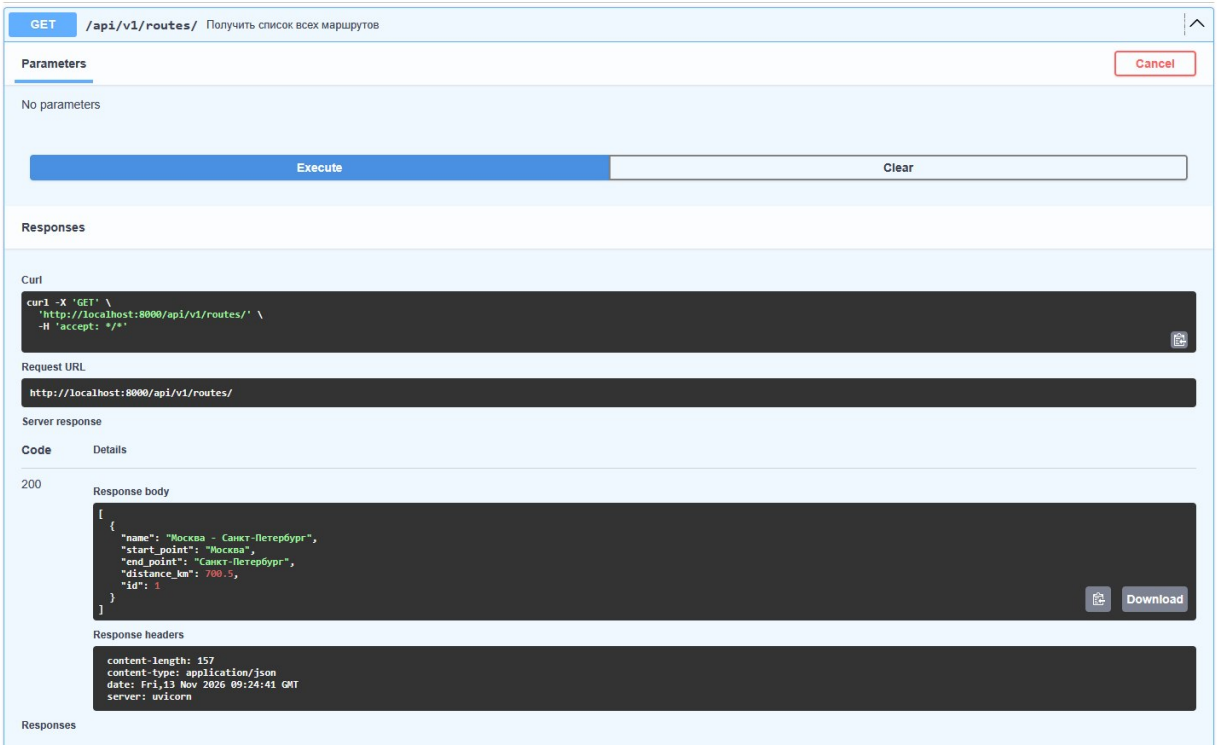
echo "✅ Структура проекта создана успешно!"
✅ Структура проекта создана успешно!

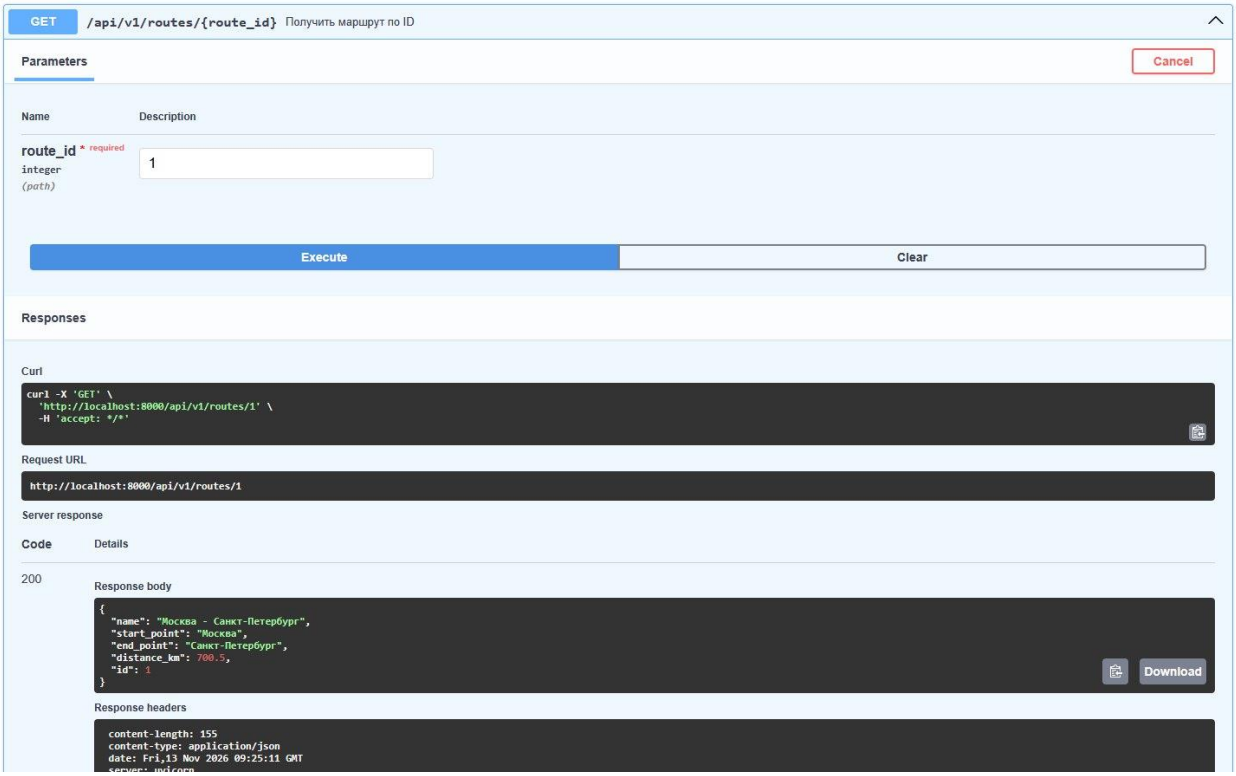
User@DESKTOP-210DI9R MINGW64 /d/ГИ-41/lab6/transport-microservice (feature/rest-api)
$
```

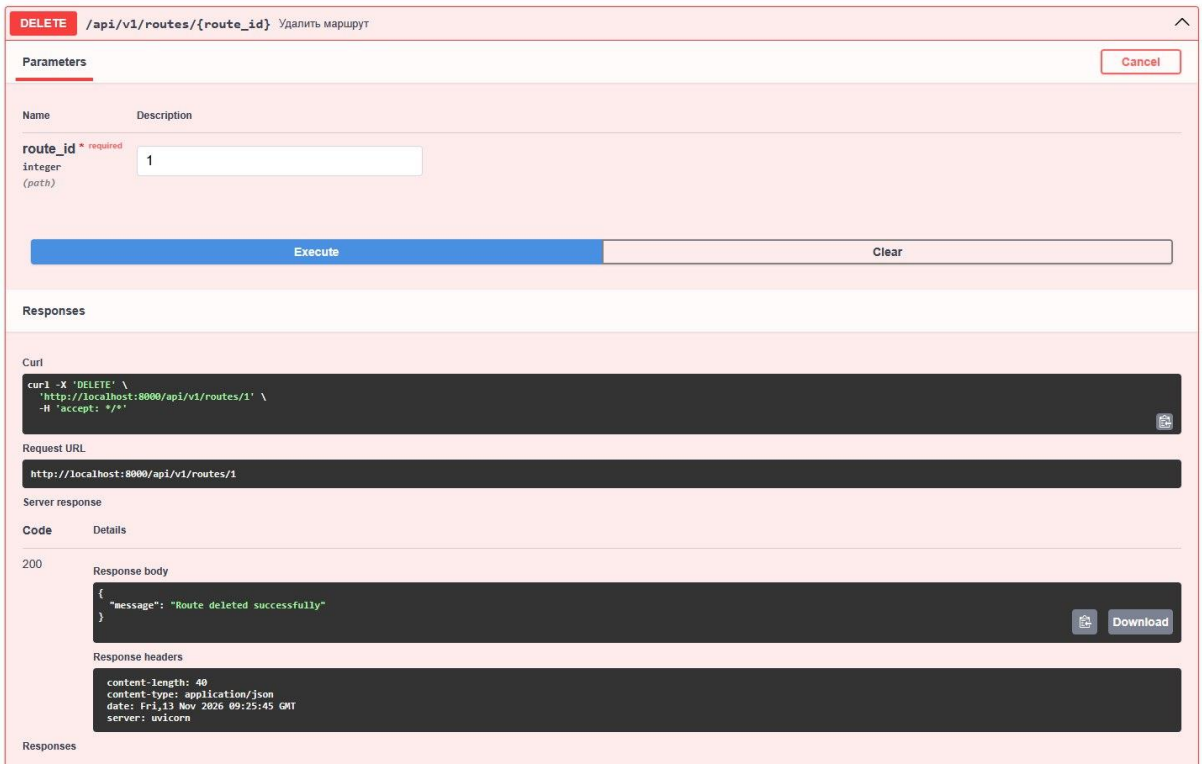

Problems
Output
Debug Console
Terminal
Ports
powershell
PS D:\ГИ-41\lab6\transport-microservice> docker-compose up
-d
=> => exporting manifest list sha256:441f94cf8f9deb1371eb09be0a1124fff986691f94306927aaa8bf56c89e0098 0.0s
=> => naming to docker.io/library/transport-microservice-transport-api:latest 0.0s
=> => unpacking to docker.io/library/transport-microservice-transport-api:latest 0.4s
=> [transport-api] resolving provenance for metadata file 0.0s
[+] Running 3/3
✓ transport-api Built 0.0s
✓ Network transport-microservice_default Created 0.0s
✓ Container transport_service Started 0.7s
PS D:\ГИ-41\lab6\transport-microservice> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
3fb68b76ae70 transport-microservice-transport-api "uvicorn app.main:ap…" 15 seconds ago Up 14 seconds 0.0.0.0:8000->8000/tcp transport_service
PS D:\ГИ-41\lab6\transport-microservice>


POST
/api/v1/routes/
Создать новый маршрут
Parameters
Cancel
Reset
No parameters
Request body
required
application/json
Edit Value
Schema
{
"name": "Москва - Санкт-Петербург",
"start_point": "Москва",
"end_point": "Санкт-Петербург",
"distance_km": 700.5
}
Execute
Clear

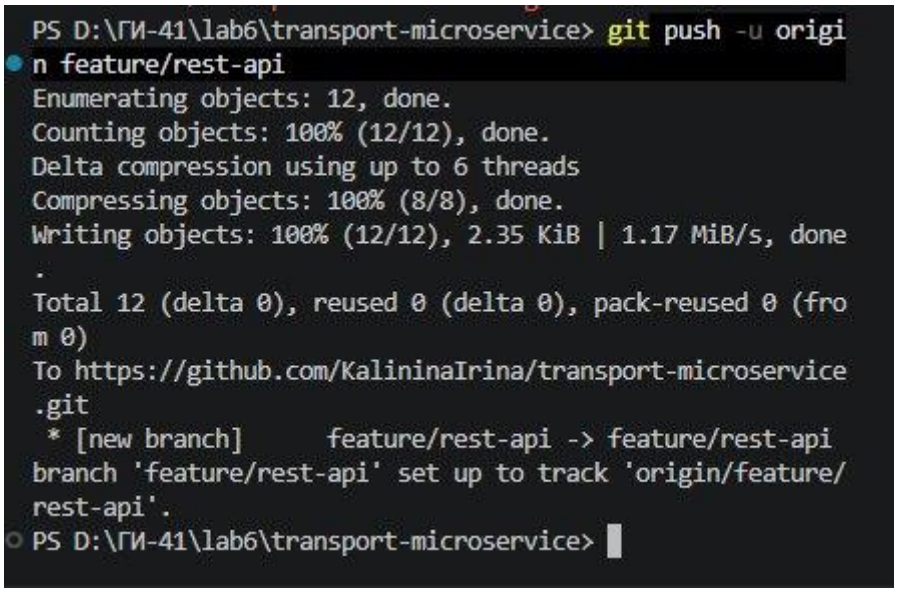

Request URL
http://localhost:8000/api/v1/routes/
Server response
Code
Details
201
Response body
{
"name": "Москва - Санкт-Петербург",
"start_point": "Москва",
"end_point": "Санкт-Петербург",
"distance_km": 700.5,
"id": 1
}
Download
Response headers
content-length: 155
content-type: application/json
date: Fri,13 Nov 2026 09:23:13 GMT
server: uvicorn
Responses
Code
Description
Links
201
Successful Response
No links
Media type
application/json
Controls Accept header.
Example Value
Schema
{
"name": "string",
"start_point": "string",
"end_point": "string",
"distance_km": 1,
"id": 0
}

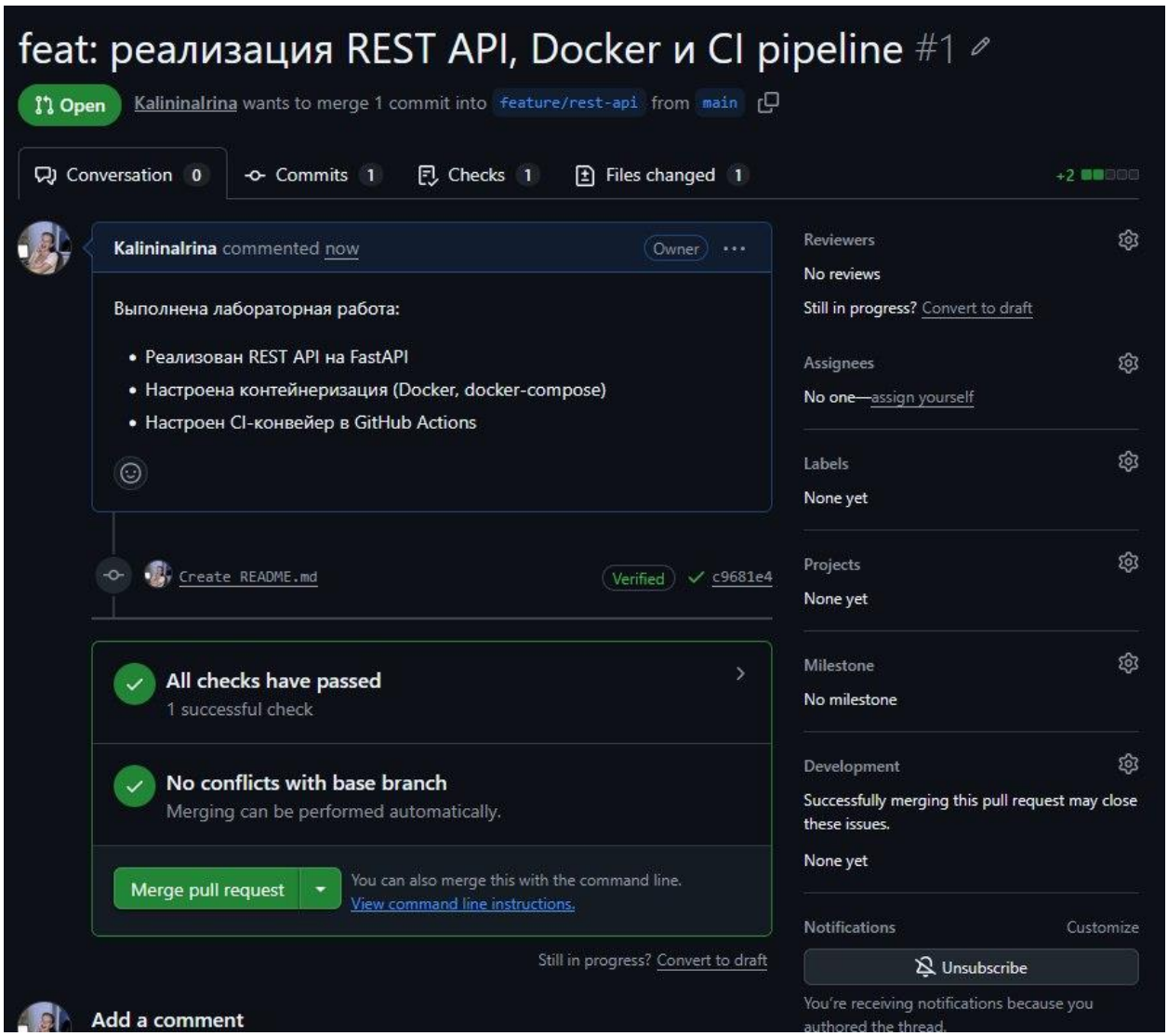

GET
/api/v1/routes/
Получить список всех маршрутов
Parameters
Cancel
No parameters
Execute
Clear
Responses
Curl
curl -X 'GET' \
'http://localhost:8000/api/v1/routes/' \
-H 'accept: */*'
Request URL
http://localhost:8000/api/v1/routes/
Server response
Code
Details
200
Response body
[
{
"name": "Москва - Санкт-Петербург",
"start_point": "Москва",
"end_point": "Санкт-Петербург",
"distance_km": 700.5,
"id": 1
}
]
Download
Response headers
content-length: 157
content-type: application/json
date: Fri,13 Nov 2026 09:24:41 GMT
server: uvicorn
Responses

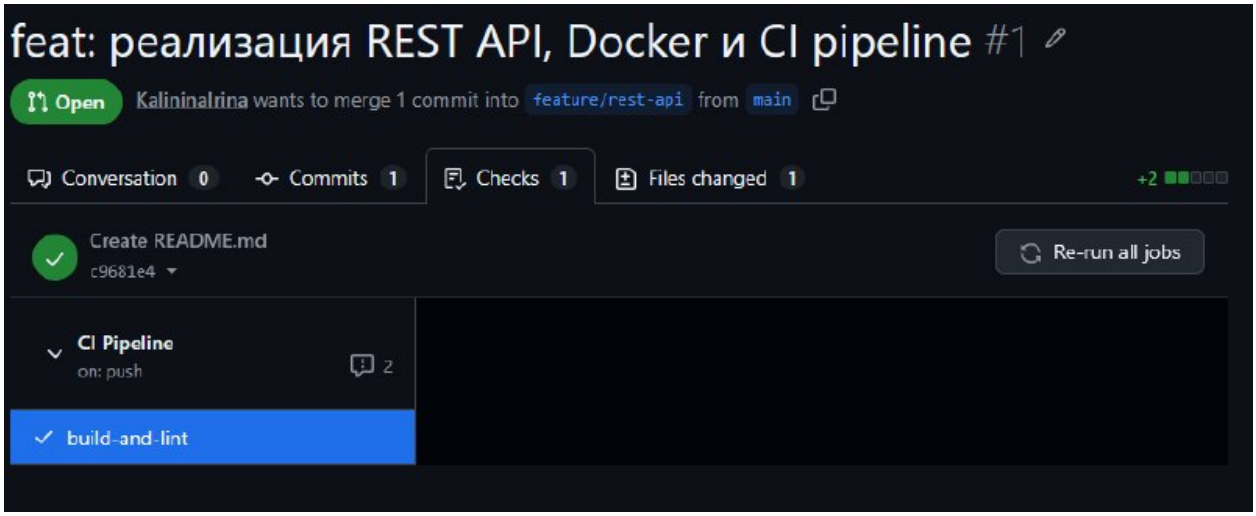

GET
/api/v1/routes/{route_id}
Получить маршрут по ID
Parameters
Cancel
Name
Description
route_id * required
integer
(path)
1
Execute
Clear
Responses
Curl
curl -X 'GET' \
'http://localhost:8000/api/v1/routes/1' \
-H 'accept: */*'
Request URL
http://localhost:8000/api/v1/routes/1
Server response
Code
Details
200
Response body
{
"name": "Москва - Санкт-Петербург",
"start_point": "Москва",
"end_point": "Санкт-Петербург",
"distance_km": 700.5,
"id": 1
}
Download
Response headers
content-length: 155
content-type: application/json
date: Fri,13 Nov 2026 09:25:11 GMT

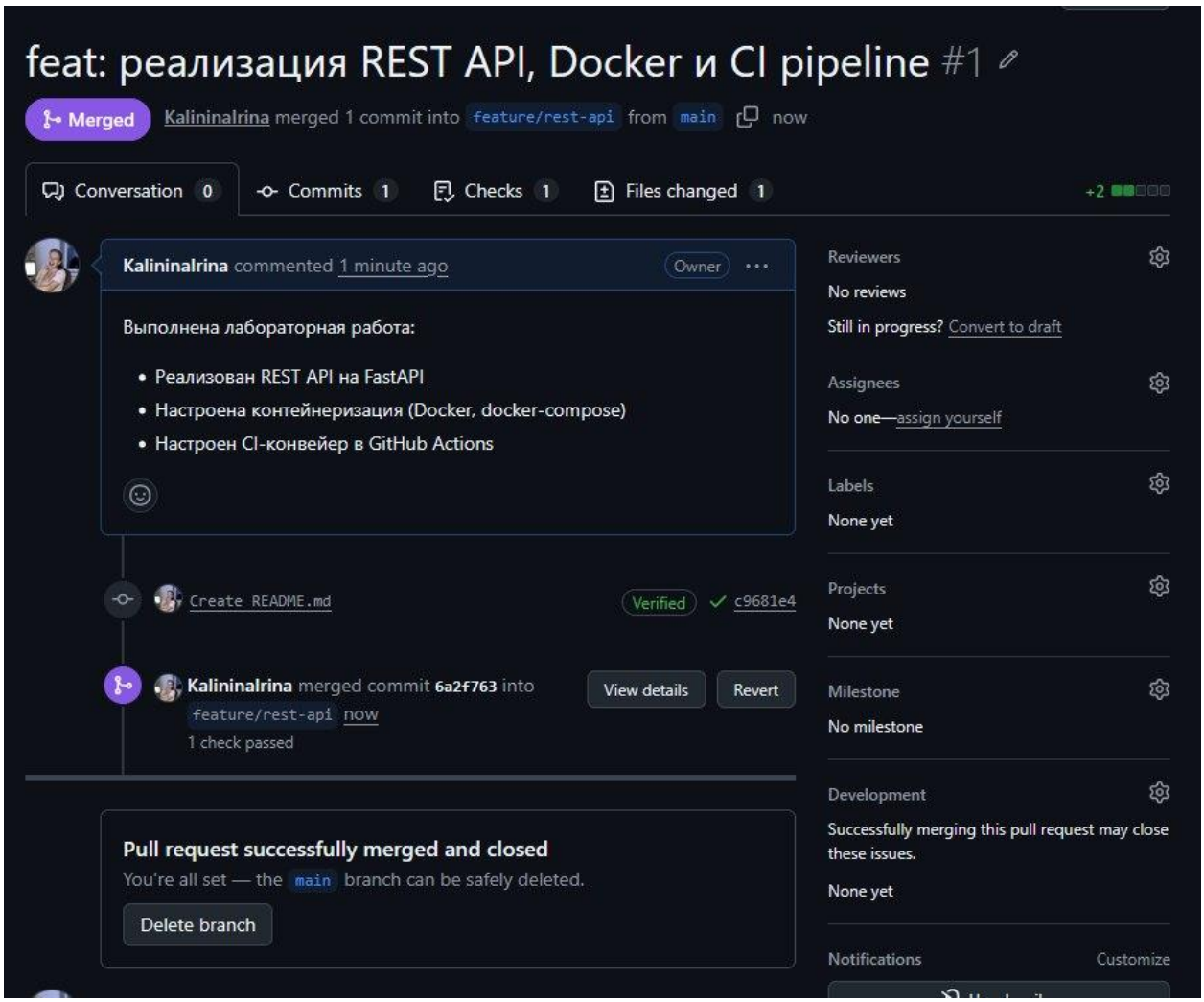

DELETE
/api/v1/routes/{route_id}
Удалить маршрут
Parameters
Cancel
Name
Description
route_id * required
integer
(path)
1
Execute
Clear
Responses
Curl
curl -X 'DELETE' \
'http://localhost:8000/api/v1/routes/1' \
-H 'accept: */*'
Request URL
http://localhost:8000/api/v1/routes/1
Server response
Code
Details
200
Response body
{
"message": "Route deleted successfully"
}
Download
Response headers
content-length: 40
content-type: application/json
date: Fri,13 Nov 2026 09:25:45 GMT
server: uvicorn
Responses


Problems
Output
Debug Console
Terminal
Ports
powershell
PS D:\ГИ-41\lab6\transport-microservice> docker-compose up
-d
=> [transport-api] resolving provenance for metadata file
0.0s
[+] Running 3/3
✓ transport-api Built
0.0s
✓ Network transport-microservice_default Created
0.0s
✓ Container transport_service Started
0.7s
PS D:\ГИ-41\lab6\transport-microservice> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
3fb68b76ae70 transport-microservice-transport-api "uvicorn app.main:ap…" 15 seconds ago Up 14 seconds 0.0.0.0:8000->8000/tcp transport_service
PS D:\ГИ-41\lab6\transport-microservice> docker-compose down
time="2026-11-13T12:26:11+03:00" level=warning msg="D:\\ГИ-41\\lab6\\transport-microservice\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 2/2
✓ Container transport_service Removed
0.7s
✓ Network transport-microservice_default Removed
0.5s
PS D:\ГИ-41\lab6\transport-microservice>


PS D:\ГИ-41\lab6\transport-microservice> git push -u origi
n feature/rest-api
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 6 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (12/12), 2.35 KiB | 1.17 MiB/s, done
.
Total 12 (delta 0), reused 0 (delta 0), pack-reused 0 (fro
m 0)
To https://github.com/KalininaIrina/transport-microservice
.git
* [new branch] feature/rest-api -> feature/rest-api
branch 'feature/rest-api' set up to track 'origin/feature/
rest-api'.
PS D:\ГИ-41\lab6\transport-microservice>


feat: реализация REST API, Docker и CI pipeline #1
Open
Kalininalrina wants to merge 1 commit into feature/rest-api from main
Conversation 0
Commits 1
Checks 1
Files changed 1
+2
Kalininalrina commented now
Owner
Выполнена лабораторная работа:
• Реализован REST API на FastAPI
• Настроена контейнеризация (Docker, docker-compose)
• Настроен CI-конвейер в GitHub Actions
Create README.md
Verified
c9681e4
All checks have passed
1 successful check
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line.
View command line instructions.
Still in progress? Convert to draft
Add a comment
Reviewers
No reviews
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Unsubscribe
You're receiving notifications because you authored the thread.


feat: реализация REST API, Docker и CI pipeline #1
Open
Kalininalrina wants to merge 1 commit into feature/rest-api from main
Conversation 0
Commits 1
Checks 1
Files changed 1
+2
Create README.md
c9681e4
Re-run all jobs
CI Pipeline
on: push
2
build-and-lint


feat: реализация REST API, Docker и CI pipeline #1
Merged
Kalininalrina merged 1 commit into feature/rest-api from main now
Conversation 0
Commits 1
Checks 1
Files changed 1
+2
Kalininalrina commented 1 minute ago
Owner
Выполнена лабораторная работа:
• Реализован REST API на FastAPI
• Настроена контейнеризация (Docker, docker-compose)
• Настроен CI-конвейер в GitHub Actions
Create README.md
Verified
c9681e4
Kalininalrina merged commit 6a2f763 into feature/rest-api now
1 check passed
View details
Revert
Pull request successfully merged and closed
You're all set — the main branch can be safely deleted.
Delete branch
Reviewers
No reviews
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize